

HealthyCooking

Project Report

Introduction

HealthyCooking is a mobile application for discovering recipes, planning meals, and tracking nutrition. It helps users maintain a balanced diet by combining personalized recommendations with practical meal planning tools, supporting users who want to eat healthier without spending excessive time researching recipes or tracking nutrition manually.

Problem Statement

Maintaining a healthy diet is difficult when recipes and meal plans are not tailored to individual needs, time, or preferences. Existing platforms often lack personalization, allergy management, or integrated nutritional tracking, making it hard for users to plan balanced meals effectively. HealthyCooking addresses this gap by combining recipe discovery, meal planning, and nutrition tracking in a single application.

Objectives

- Provide wholesome recipe recommendations based on user preferences and dietary goals.
- Account for calories, available ingredients, and food allergies when suggesting meals.
- Offer a simple, user-friendly interface for easy interaction.
- Use AI to generate new recipes or adjust existing ones based on user needs.
- Connect to a nutrition and calorie database for accurate nutrient information.
- Support meal scheduling and reminders to help users follow their diet plans.
- Allow users to search recipes by available preparation time.
- Enable users to share recipes and interact with a community of other users.

Key Features

- Recipe discovery with filters for cuisine, dietary restrictions, preparation time, and more.
- Personalized recipe recommendations based on user preferences and goals.
- Smart ingredient substitutions when items are missing.
- Weekly meal planner with full control to adjust generated plans.
- Integration with nutritional databases to calculate calories and macronutrients.
- Recipe and community sharing with comments and ratings.
- User profile management with dietary goals, allergies, and preferences.

- Analytics dashboard with visual reports on nutrition and progress.

System Architecture

The system follows a three-layer architecture:

- Presentation Layer: handles all user interaction. Built with SwiftUI for iOS and XML-based layouts for Android.
- Application and Logic Layer: handles authentication, profile management, recipes, and meal planning. Built with Swift for iOS and Kotlin for Android.
- Database Layer: responsible for storing, retrieving, updating, and deleting system data securely, using a SQL-based database.

The application is designed as a native mobile product for iOS and Android, with cloud-based storage to reduce dependence on device hardware and keep data available across sessions.

Development Approach

The project follows an incremental delivery process, with documentation and features built up in stages across defined phases: project planning, requirements analysis, system design, prototype development, and testing. Each phase produces a reviewed deliverable before the next phase begins.

Design and prototyping rely on Figma and Canva, documentation is written in Microsoft Word following IEEE and university templates, and task scheduling is tracked through a shared calendar. This structured approach keeps the scope clear and the deliverables consistent from one phase to the next.

Testing

The application is verified through functional and usability testing at each stage of development. Test cases cover core features such as recipe search, meal planning, and nutrition tracking, with a focus on accuracy, performance, and a smooth user experience. Issues found during testing are corrected before moving to the next development phase.

Conclusion

HealthyCooking brings recipe discovery, meal planning, and nutrition tracking together in one application, helping users build healthier eating habits without extra effort. Its layered architecture and native mobile design keep the system responsive and secure, while its AI-assisted recommendations make healthy eating easier to sustain over time.